

# High-Frequency US in Hepatobiliary Imaging

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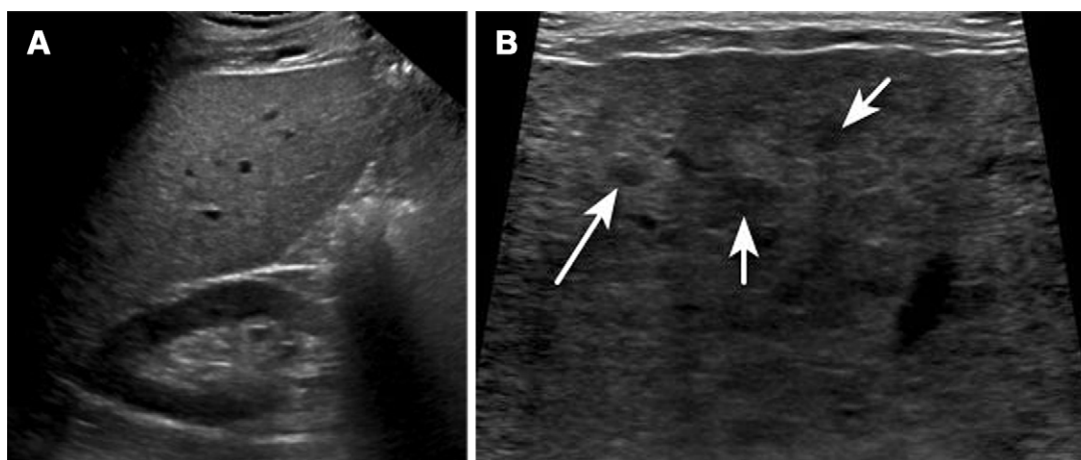
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See the slide presentation [here](#).

High-frequency (HF;  $\geq 12$  MHz) US can be used in routine hepatobiliary imaging examinations. The accompanying slide presentation demonstrates our use of HF US to evaluate the liver surface and parenchyma, peripheral focal liver lesions, gallbladder wall, and bile ducts to make diagnoses that were either questionable or not appreciated at standard lower-frequency US. We discuss our routine HF US protocol, correlate HF US and histologic findings, and highlight the modality's shortfalls.

HF US is an established method to evaluate the contour and margins of the liver for diagnosis of cirrhosis. While it cannot usually penetrate the full liver, HF US allows improved resolution of the superficial hepatic parenchyma. Thus, with diffuse liver disease, subtle findings are often more easily appreciated, especially for less experienced interpreters. In addition to several static HF images, our protocol requires HF US cine clips of the right and left lobes, allowing the interpreter to make an independent assessment of the hepatic contour and texture of the parenchyma. Instead of the commonly used description of coarse hepatic parenchyma seen on low-frequency images, HF US images directly depict increased numbers of hyperechoic reticular echoes felt to represent fibrosis and distinct 1–3-mm regenerating nodules that define cirrhosis (Fig 1).

Although US is typically the initial imaging modality used to evaluate the gallbladder, its high sensitivity for acute cholecystitis, especially as compared with that of CT, has been questioned in recent years. Since the gallbladder is relatively superficial, it is well suited to HF US. We have found that subtle signs of acute cholecystitis are also easily appreciated, including slight wall thickening, tensile shape, small amounts of fluid, and adjacent echogenic fat. Demonstration of these overt findings is much more useful for diagnosis than the subjective sonographic Murphy sign (Fig 2). The gallbladder wall can be scrutinized for subtle defects or trace amounts of gas to make the diagnosis of gangrenous cholecystitis. In addition to gallbladder wall edema in acute cholecystitis, HF US can help more confidently differentiate other benign gallbladder conditions from malignant gallbladder conditions. In focal adenomyomatosis, one may directly visualize the dilated crystal-filled Rokitsansky-Aschoff sinuses and avoid the need for follow-up or alternative imaging to exclude gallbladder carcinoma.

While lower-frequency imaging is excellent to evaluate bile ducts and vessels in the central liver, the smaller branches in the periphery of the liver are well visualized at HF US. Thus, one may more easily distinguish between periportal edema and biliary sludge and differentiate benign from malignant portal vein



**Figure 1.** Cirrhosis at HF abdominal US in a 79-year-old man with a history of diabetes mellitus and hypertension who underwent imaging because of abnormal liver function test results. (A) US image obtained at low frequency using a C5–1-MHz transducer shows increased echogenicity of the liver compared with the right kidney and a smooth contour, consistent with steatosis. (B) US image obtained using a L12–3-MHz transducer shows a wavy slightly nodular contour of the liver surface and numerous hypoechoic regenerating nodules (arrows) interspersed with wavy echogenic bands of fibrosis, indicative of cirrhosis. These findings are only appreciated at HF US.



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 Abbreviation: HF = high frequency

## TEACHING POINTS

- Beware of making or excluding a diagnosis of cirrhosis based solely on the liver contour; corroborate with confirmatory textural findings at HF US.
- The tensile shape and subtle wall thickening of an inflamed gallbladder in acute cholecystitis can be overlooked at routine low-frequency US.
- Direct visualization of dilated Rokitsansky-Aschoff sinuses using HF imaging can help distinguish focal adenomyomatosis from a gallbladder mass.

thrombosis. Using HF US helps delineate the various causes of punctate and linear echogenic foci, including pneumobilia, von Meyenburg complexes, and hepatic cysts.

Last, we provide examples of situations in which HF US is an impediment rather than an improvement, including severe steatosis and large volumes of air or gas.

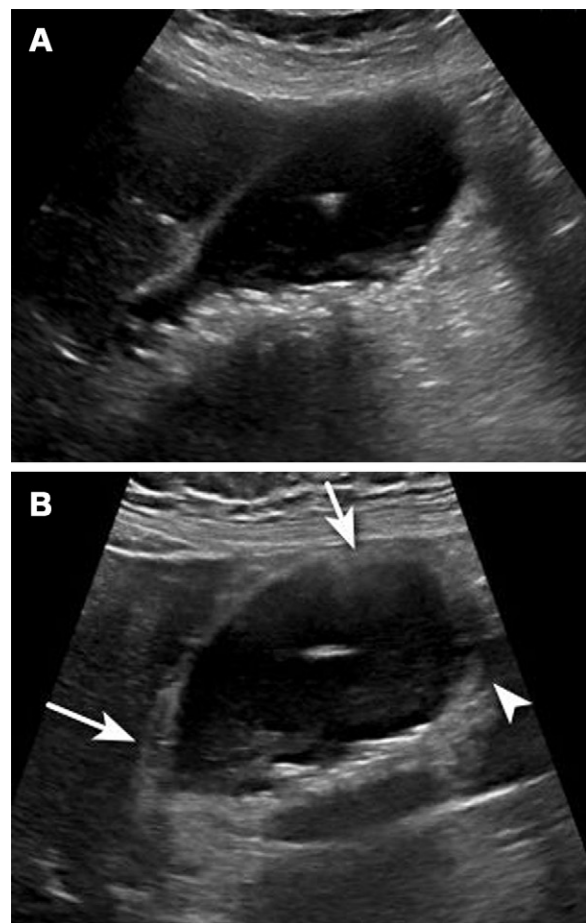
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## Suggested Readings

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**Figure 2.** Acute cholecystitis findings at HF US in a 66-year-old woman with right upper quadrant pain. US image (A) shows findings that were interpreted as cholelithiasis without wall thickening or sonographic Murphy sign. Because of the patient's pain, abdominal CT (not shown) was performed and demonstrated findings of acute cholecystitis, including a distended gallbladder with a thickened edematous wall. The abdominal CT findings prompted a review of the HF gallbladder images from the US acquisition. HF US image (B) shows a thick edematous wall (arrows), pericholecystic fluid (arrowhead), and a tensile shape, in addition to calculi and sludge.

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